

PRACTICAL NO. 6

AIM: Considered there are N philosophers seated around a circular table with one chopstick between each pair of philosophers. There is one chopstick between each philosopher. A philosopher may eat if he can pick up the two chopsticks adjacent to him. one chopstick may be picked up by any one of its adjacent followers but not both. Write a program to solve the problem using process synchronization technique.

CODE:

```
MINGW64/c/Users/airat/OneDrive/Desktop
GNU nano 8.7 dining.c
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define N 5 // Number of philosophers

sem_t chopstick[N];
sem_t room; // To prevent deadlock

void* philosopher(void* num) {
    int id = *(int*)num;

    printf("Philosopher %d is thinking\n", id);
    sleep(1);

    sem_wait(&room); // Enter room

    sem_wait(&chopstick[id]); // Pick left chopstick
    printf("Philosopher %d picked left chopstick\n", id);

    sem_wait(&chopstick[(id + 1) % N]); // Pick right chopstick
    printf("Philosopher %d picked right chopstick\n", id);

    printf("Philosopher %d is eating\n", id);
    sleep(2);

    sem_post(&chopstick[id]); // Release left
    sem_post(&chopstick[(id + 1) % N]); // Release right
    printf("Philosopher %d finished eating and released chopsticks\n", id);

    sem_post(&room); // Leave room

    return NULL;
}

int main() {
    pthread_t thread[N];
    int phil[N];

    sem_init(&room, 0, N-1); // Allow max N-1 philosophers
    for(int i = 0; i < N; i++)
        sem_init(&chopstick[i], 0, 1);
}
```

OUTPUT:

```
MINGW64/c/Users/airat/OneDrive/Desktop
Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ nano dining.c

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ gcc dining.c -o dining

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$ ./dining
Philosopher 0 is thinking
Philosopher 3 is thinking
Philosopher 2 is thinking
Philosopher 1 is thinking
Philosopher 4 is thinking
Philosopher 2 picked left chopstick
Philosopher 4 picked left chopstick
Philosopher 4 picked right chopstick
Philosopher 4 is eating
Philosopher 1 picked left chopstick
Philosopher 3 picked left chopstick
Philosopher 4 finished eating and released chopsticks
Philosopher 0 picked left chopstick
Philosopher 3 picked right chopstick
Philosopher 3 is eating
Philosopher 2 picked right chopstick
Philosopher 2 is eating
Philosopher 3 finished eating and released chopsticks
Philosopher 2 finished eating and released chopsticks
Philosopher 1 picked right chopstick
Philosopher 1 is eating
Philosopher 1 finished eating and released chopsticks
Philosopher 0 picked right chopstick
Philosopher 0 is eating
Philosopher 0 finished eating and released chopsticks

All philosophers have finished eating.

Faiz@PUCKC3F09N9DT MINGW64 ~/OneDrive/Desktop (master)
$
```

